

Overview of U.S. COVID-19 vaccine safety surveillance systems[☆]

Julianne Gee^{a,*}, Tom T. Shimabukuro^a, John R. Su^a, David Shay^a, Margaret Ryan^b, Sridhar V. Basavaraju^a, Karen R. Broder^a, Matthew Clark^c, C. Buddy Creech^d, Francesca Cunningham^e, Kristin Goddard^f, Harrison Guy^c, Kathryn M. Edwards^d, Richard Forshee^g, Tanya Hamburger^a, Anne M. Hause^a, Nicola P. Klein^f, Ian Kracalik^a, Chris Lamer^c, David A. Loran^b, Michael M. McNeil^a, Jay Montgomeryⁱ, Pedro Moro^a, Tanya R. Myers^a, Christine Olson^a, Matthew E. Oster^{h,j}, Andrea J. Sharma^a, Ryan Schupbach^c, Eric Weintraub^a, Brett Whitehead^c, Steven Anderson^g

^a National Center for Emerging Zoonotic Infectious Diseases, Centers for Disease Control and Prevention (CDC), Atlanta, GA, United States

^b Defense Health Agency, Immunization Healthcare Division, San Diego, CA, United States

^c Indian Health Service (IHS), IHS National Pharmacy & Therapeutics Committee, Durango, CO, United States

^d Vanderbilt Vaccine Research Program, Vanderbilt University Medical Center and School of Medicine, Nashville, TN, United States

^e Department of Veterans Affairs, Veterans Affairs Center for Medication Safety - Pharmacy Benefit Management Services, Hines, IL, United States

^f Kaiser Permanente Vaccine Study Center, Kaiser Permanente Northern California, Oakland, CA, United States

^g Office of Biologics and Pharmacovigilance, Center for Biologics Evaluation and Research, US Food and Drug Administration, Silver Spring, MD, United States

^h National Center for Birth Defects and Developmental Disabilities, CDC, Atlanta GA, United States

ⁱ Defense Health Agency, Immunization Healthcare Division, Bethesda, MD, United States

^j Emory University School of Medicine, Children's Healthcare of Atlanta, Atlanta, GA, United States

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ABSTRACT

The U.S. COVID-19 vaccination program, which commenced in December 2020, has been instrumental in preventing morbidity and mortality from COVID-19 disease. Safety monitoring has been an essential component of the program. The federal government undertook a comprehensive and coordinated approach to implement complementary safety monitoring systems and to communicate findings in a timely and transparent way to healthcare providers, policymakers, and the public. Monitoring involved both well-established and newly developed systems that relied on both spontaneous (passive) and active surveillance methods. Clinical consultation for individual cases of adverse events following vaccination was performed, and monitoring of special populations, such as pregnant persons, was conducted. This report describes the U.S. government's COVID-19 vaccine safety monitoring systems and programs used by the Centers for Disease Control and Prevention, the U.S. Food and Drug Administration, the Department of Defense, the Department of Veterans Affairs, and the Indian Health Service. Using the adverse event of myocarditis following mRNA COVID-19 vaccination as a model, we demonstrate how the multiple, complementary monitoring systems worked to rapidly detect, assess, and verify a vaccine safety signal. In addition, longer-term follow-up was conducted to evaluate the recovery status of myocarditis cases following vaccination. Finally, the process for timely and transparent communication and dissemination of COVID-19 vaccine safety data is described, highlighting the responsiveness and robustness of the U.S. vaccine safety monitoring infrastructure during the national COVID-19 vaccination program.

1. Introduction

The emergence of the SARS-CoV-2 virus in 2019 and the subsequent

COVID-19 pandemic has had a devastating impact on public health worldwide. In the United States, there were more than 103 million COVID-19 cases and 1.1 million deaths [1], with an estimated nationwide

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* Corresponding author at: National Center for Emerging Zoonotic Infectious Diseases, Centers for Disease Control and Prevention (CDC), 1600 Clifton RD, Atlanta, GA 30329, United States.

E-mail address: dzg2@cdc.gov (J. Gee).

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economic loss of over \$14 trillion anticipated by the end of 2023 [2,3]. COVID-19 vaccines have played a key role in preventing morbidity and mortality and mitigating societal disruption from the COVID-19 pandemic [4]. The urgency to acquire safe and effective COVID-19 vaccines accelerated the development and use of novel vaccine manufacturing technologies for human vaccines. Prior to the COVID-19 pandemic, some of these technologies had been in limited use or considered experimental for human vaccines [5]. Expedited regulatory pathways and the incorporation and widespread use of these manufacturing technologies (e.g., mRNA-based techniques) spurred rapid vaccine production on an international scale and enabled an expedited timeline for authorization and program implementation. COVID-19 vaccines were extensively evaluated for efficacy and safety in clinical trials involving tens of thousands of volunteer study subjects; however, even large clinical trials, such as the COVID-19 vaccine pre-authorization clinical trials, were limited in their ability to detect rare adverse events (AE). While accelerated [6], the regulatory process for U.S. COVID-19 vaccines fulfilled the rigorous requirements for U.S. Food and Drug Administration (FDA) Emergency Use Authorization (EUA) and subsequently, for some vaccines, licensure.

Obtaining timely and accurate safety data during the largest public vaccination program in U.S. history required coordination across multiple U.S. government agencies. Recurring meetings to discuss planning and implementation efforts for vaccine safety monitoring and to conduct timely review of safety findings as data were collected and analyzed were implemented. Federal partners included the Centers for Disease Control and Prevention (CDC), the FDA, the Department of Veterans Affairs (VA), the Department of Defense (DoD), and the Indian Health Service (IHS). In advance of the national vaccination program, CDC and FDA developed a core list of adverse events of special interest (AESI) (also referred to as ‘pre-specified outcomes’) for enhanced safety monitoring (Table 1). These AESIs were chosen based on AEs of historical concern for vaccine safety (e.g., Guillain-Barré syndrome [GBS]), serious allergic AEs known to be associated with vaccination (e.g., anaphylaxis), theoretical safety concerns and outcomes based on certain COVID-19 disease complications, biologic plausibility, and findings from the COVID-19 vaccine clinical trials. Other AESIs were added for monitoring based on surveillance findings as the vaccination program progressed. Use of pre-specified AESI surveillance outcomes, application

of standardized case definitions when available (including Brighton Collaboration case definitions) [7], and interagency collaboration on surveillance protocol development ensured a measure of consistency across safety evaluations. Monitoring efforts through many of the federal systems focused on detecting safety signals for AEs. The Council for International Organizations of Medical Sciences defines a signal as “Information...from one or multiple sources..., which suggest a new potential casual association, or a new aspect of a known association between an intervention and an event, or set of related events, either adverse or beneficial, that is judged to be of sufficient likelihood to justify verifactory action” [8].

As of August 2023, COVID-19 vaccine safety data continue to accrue from a network of U.S. government vaccine safety monitoring systems. (Fig. 1, Table 2). These systems included spontaneous reporting (passive surveillance) systems, such as the Vaccine Adverse Event Reporting System (VAERS) [9], a national CDC and FDA co-managed program that accepts reports from vaccine manufacturers, healthcare providers, and the public, including reports originating from VA’s Adverse Event Reporting System (ADERS), DoD’s Defense Health Agency’s Immunization Health Division (DHA IHD), and IHS healthcare facilities [10]. Active surveillance systems, comprised primarily of large-linked database systems, used near real-time sequential monitoring methods [11] and included the CDC’s Vaccine Safety Datalink (VSD) [12], the FDA’s Biologics Effectiveness and Safety System (BEST) [13], the FDA’s and Centers for Medicare and Medicaid Services (CMS) collaboration on analyses of data [14], and the VA’s analyses of VA electronic health record (EHR) data [15]. Additionally, DHA conducted population-based surveillance using the Defense Medical Surveillance System (DMSS) [16]. CDC’s Clinical Immunization Safety Assessment (CISA) Project and DHA Immunization Healthcare Division (IHD) also conducted clinical consultations for complex vaccine safety cases at the request of U.S. healthcare providers and consulted on surveillance activities [17,18]. Several new COVID-19-specific vaccine safety programs were implemented to supplement the existing systems, including CDC’s v-safe [19], IHS’s National Pharmacy and Therapeutics Committee (NPTC) sentinel site surveillance program [20], CDC’s COVID-19 Vaccine Pregnancy Registry [21], and a CDC enhanced surveillance project to obtain information on longer-term outcomes in patients experiencing

Table 1
Core list of adverse events of special interest monitored by U.S. federal agencies.

Outcome	VAERS	VSD	VA	DMSS	DHA IHD	CMS	BEST
Acute myocardial infarction	x	x	x	x	–	x	x
Anaphylaxis	x	x	x	x	x	x	x
Appendicitis	x	x	x	x	x	x	x
Bell’s palsy	x	x	x	x	x	x	x
Disseminated intravascular coagulation	x	x	x	x	–	x	x
Encephalomyelitis/Encephalitis	–	–	–	x	–	x	x
Encephalitis	x	x	x	–	x	–	–
Encephalomyelitis	x	x	x	–	x	–	–
Guillain-Barré syndrome	x	x	x	x	x	x	x
Immune thrombocytopenic purpura	–	x	x	x	x	x	x
Multisystem Inflammatory Syndrome in Adults	x	x	x	x	–	x	x
Multisystem Inflammatory Syndrome in Children	x	x	–	–	–	–	x
Myocarditis / pericarditis	x	x	x	x	x	x	x
Narcolepsy / cataplexy	x	x	x	x	x	x	x
Stroke	x	–	x	–	x	–	–
Non-hemorrhagic stroke	–	x	x	–	–	x	x
Hemorrhagic stroke	–	x	x	–	–	x	x
Thrombosis with thrombocytopenia syndrome and/or cerebral venous sinus thrombosis	x	x	–	x	–	x	x
Transverse myelitis	x	x	x	x	x	x	x
Venous thromboembolism	x	x	x	x	x	x	x
Pulmonary embolism	–	x	x	x	x	x	x
Deep vein thrombosis	–	–	x	x	–	x	x

Note: “x” indicates a health outcome that was monitored in the vaccine safety system; health outcomes in each system varied depending on age group and vaccine. The Indian Health Service is not included in this table, as no pre-specified outcomes in the monitoring systems.

Abbreviations: VAERS, Vaccine Adverse Event Reporting System; VSD, Vaccine Safety Datalink; VA, Department of Veterans Affairs Warehouse; DMSS, Defense Medical Surveillance System; DHA IHD, Defense Health Agency Immunization Health Division; CMS, Centers for Medicare & Medicaid Services; BEST, Biologics Effectiveness and Safety System.

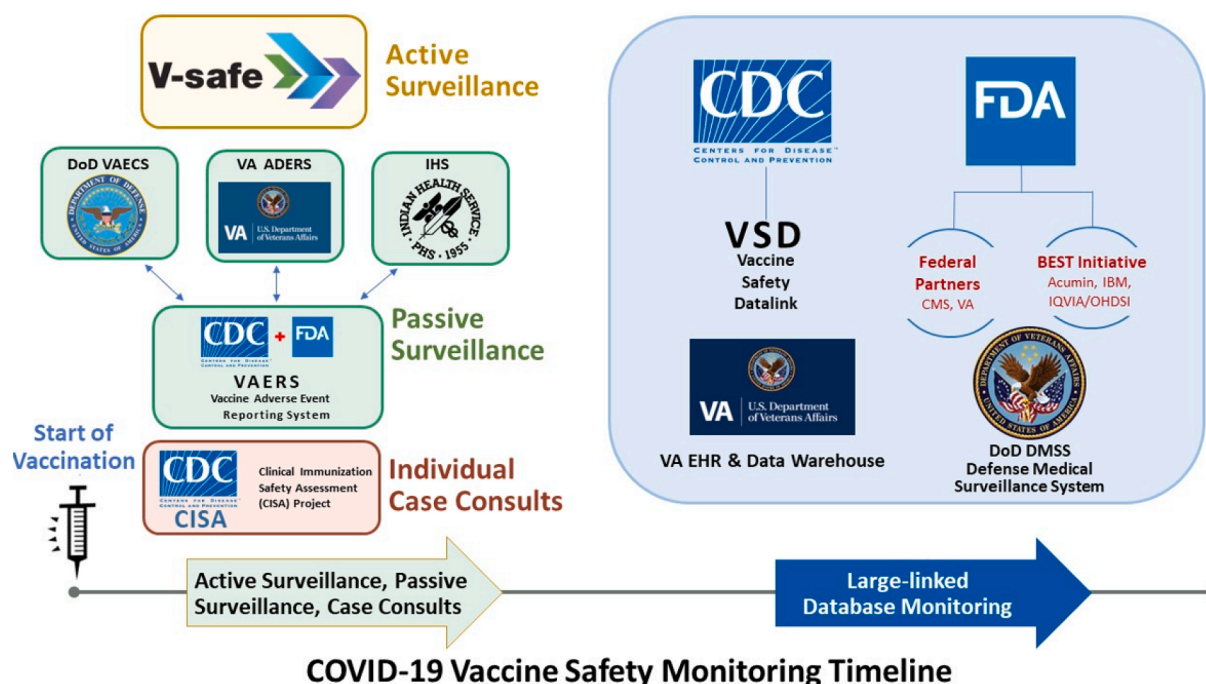


Fig. 1. U.S. federal COVID-19 vaccine safety monitoring systems and programs.

myocarditis after mRNA COVID-19 vaccination [22].

We describe the U.S. federal COVID-19 vaccine safety monitoring systems and programs (Table 2). Using the example of myocarditis following mRNA COVID-19 vaccines, we illustrate how several of these systems were used to rapidly evaluate the safety of COVID-19 vaccines. We also describe the channels for communication and dissemination of COVID-19 vaccine safety data. Finally, we discuss the importance of coordinated, federal vaccine safety efforts and lessons learned.

2. Spontaneous Reporting Systems

Spontaneous reporting systems, also known as passive surveillance systems, are the backbone of post-authorization and post-licensure vaccine safety monitoring. Under selected circumstances, reporting of certain AEs following vaccination is mandated by regulatory and public health authorities (e.g., vaccine manufacturer and healthcare provider reporting requirements). These systems also depend on voluntary reporting by healthcare providers, public health officials, and the public. For COVID-19 vaccine safety monitoring, the United States relied on VAERS to provide initial safety data and serve as the main system for early signal detection. According to code of federal regulations, reports to VAERS are classified as 'serious' if any of the following are documented in the AE description: hospitalization, prolongation of hospitalization, life-threatening illness, permanent disability, congenital anomaly or birth defect, or death [23]. Strengths of spontaneous reporting systems include the large populations under surveillance, which allows for rapid signal detection and detection of rare AEs, and the rapid acquisition of safety data following program implementation. Limitations include the voluntary nature of reporting, reporting biases, inconsistency in the quality and completeness of reports, and lack of comparison groups. Additionally, accurate and timely information on vaccine doses administered is often not available, thus preventing the estimation of adverse event reporting rates. Using data from spontaneous reporting systems alone it is usually not possible to establish a causal association between a reported adverse event and vaccination; however, spontaneous reporting systems are adept at rapidly detecting potential safety signals that may warrant further assessment.

2.1. Vaccine Adverse Event Reporting System (VAERS)

Established in 1990, VAERS serves as the nation's early warning system for potential vaccine safety problems, especially for new vaccines during introduction [9]. VAERS received around 40,000–50,000 reports annually in the years immediately prior to the COVID-19 pandemic. However, since the COVID-19 vaccination program began in December 2020, VAERS has received over 1 million U.S. reports for COVID-19 vaccines alone. AE reporting requirements mandated by FDA and CDC, the large volume of COVID-19 vaccines administered in a relatively short period of time, and increased public attention to various safety issues from the media and awareness of the vaccination program likely contributed to this marked increase in VAERS reporting [24].

In preparation for COVID-19 vaccine safety monitoring, CDC and FDA developed the VAERS COVID-19 Vaccines Standard Operating Procedures (SOP) [25], which included descriptions of surveillance activities and analytic methods, including the monitoring of COVID-19 vaccine safety during pregnancy and a list of pre-specified AESIs for enhanced monitoring efforts; these efforts included requesting additional documents, abstraction of medical records, and outcome verification based on standardized case definitions. The SOP also included guidance on procedures to be followed for interagency coordination and collaboration. The availability of data on the number of vaccine doses administered for calculating AE reporting rates greatly enhanced the value of VAERS surveillance beyond that for other routine vaccine safety surveillance, which typically relies on data on vaccine doses distributed from the manufacturers as denominators. An example of the value added was the rapid detection and assessment of the estimated risk for thrombosis with thrombocytopenia syndrome (TTS) associated with receipt of the Janssen/J&J (Ad26.CoV2.S) vaccine within weeks of its introduction in the United States [26].

2.2. Vaccine Adverse Event Reporting System for Indian Health Service

COVID-19 vaccine safety surveillance in IHS was designed to support reporting of AEs to VAERS within the American Indian and Alaska Native populations. In November 2020, IHS collaborated with CDC to incorporate new functionality into the VAERS report form, allowing

Table 2
Characteristics of U.S. government COVID-19 vaccine safety monitoring systems and programs.

System	Managing federal agency	Data source	Population under surveillance	Characteristics	Strengths specific to COVID-19 vaccine safety monitoring	Limitations specific to COVID-19 vaccine safety monitoring
Spontaneous reporting (passive surveillance) systems						
VAERS	CDC and FDA	- Primarily voluntary reports - Healthcare providers and vaccine manufacturers are required to report certain AEs after COVID-19 vaccination - Online and electronic reporting processes exist	Entire United States (also includes reports from foreign sources)	- Serves as an early vaccine safety warning system - Performs enhanced monitoring of reported AESIs and verification of selected AESI reports - Reviews COVID-19 vaccine safety reports among pregnant persons	- Accepts reports from anyone - Data are made publicly available - Able to calculate reporting rates using vaccine doses administered from the CDC COVID Data Tracker - Provides timely data to inform COVID-19 policy recommendation	- Variable data quality and completeness - Reporting biases (underreporting and stimulated reporting) - Availability and timeliness of access to medical records during follow-up may be limited - Generally cannot assess causality for many outcomes
IHS VAERS	IHS	- Healthcare providers from IHS facilities are required to report certain AEs after COVID-19 vaccination - Online reporting processes exist	AI/AN populations	- VAERS reports originating from IHS federal, tribal, and Urban Indian Organization facilities	- Contributes data to VAERS surveillance overall - Allows for specific analyses of AI/AN populations	- Subject to the same limitations of passive surveillance
I-STAR	IHS	- Web-based digital platform that collects vaccine safety data on employees and patients, and safety events at IHS facilities	AI/AN populations	- Platform allowing for the early detection of employee-related COVID-19 vaccine AEs	- Facilitates reporting to VAERS	- Limited data from tribal and Urban Indian Organization facilities - Subject to the limitations of passive surveillance
VA ADERS	VA	- Web-based system for VA healthcare providers to report AEs following drugs and vaccines	U.S. Veteran population and VA employees receiving care at VA healthcare facilities	- Drug and vaccine safety reporting for the veteran population - Enhanced monitoring for AESIs	- Weekly reporting of non-serious and serious AEs - Facilitates reporting to VAERS	- Subject to the limitations of passive surveillance - Medical record review limited to only reports of anaphylaxis and death
DHA IHD	DoD	VAERS reports in patients indicating U.S. military or DoD affiliation	Military beneficiaries	- Conducts reviews of cases including clinical follow-up and appropriate case management - Serves as an early warning system for AEs in the U.S. military population	- Bi-directional relationship with VAERS - Able to calculate reporting rates and make risk comparisons within the U.S. military population	- Subject to the limitations of passive surveillance
Active surveillance systems						
VSD	CDC	Electronic health record data from 13 integrated healthcare organizations across the United States	~15.5 million health plan members annually, with an annual birth cohort of ~115,000 births	- Near real-time monitoring of AESIs - AESIs monitored among all age groups as appropriate - Well-defined insured population - COVID-19 vaccine safety evaluations conducted in pregnant persons	- Serves as a signal detection and signal assessment system - High quality and complete data available from EHR system - Uses a concurrent comparator methodology to minimize the effects of confounding - Ability to rapidly conduct medical record review to verify outcomes	- Represents an insured population - Findings may not be generalizable to the broader U.S. population - May not be statistically powered to rapidly detect very rare AEs
BEST	FDA	Large-scale administrative claims data from three health plans	~75 million health plan members, >100 million patients covered by EHRs	- Near real-time monitoring of AESIs - Analyses employ a historical comparison - Monitored AESIs among persons 12–64 years of age - Administrative claims data links to state immunization data to capture COVID-19 vaccinations	- Large surveillance population - Infrastructure includes powerful analytic methods such as on-demand analytics - A large network of partners who provide subject matter expertise - Conducts self-controlled analyses and medical record reviews to assess statistical signals detected in real-time surveillance	- Limited ability to control for confounding for near real-time monitoring - Data lag of >30 days for claims data

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Table 2 (continued)

System	Managing federal agency	Data source	Population under surveillance	Characteristics	Strengths specific to COVID-19 vaccine safety monitoring	Limitations specific to COVID-19 vaccine safety monitoring
FDA-CMS collaboration	FDA	Administrative Medicare data	~60 million people ages ≥ 65 years	- Near real-time monitoring of AESIs - Analyses employ a historical comparison - Includes enrollment and claims data containing demographic and longitudinal information on healthcare encounters	- Large population of older Americans ages ≥ 65 years - Conducts self-controlled analyses and medical record reviews to assess statistical signals detected in real-time surveillance	- Limited ability to control for confounding for near real-time monitoring
VA EHR system- Vaccine Rapid Cycle Analysis	VA	Electronic health records and administrative data from the VA healthcare system	~9 million veterans enrolled in the VA healthcare system	- Near real-time monitoring of AESIs - Analyses employs a historical comparison	- Ability to conduct medical record review to verify outcomes - Large data source for older adults	- Health information for medical encounters outside the VA healthcare system may not be captured - Full evaluations included only veterans who consistently use the VA healthcare system - Data may not be generalizable to populations outside the VA healthcare system
DMSS	DoD	U.S. military and DoD electronic health record systems	Military beneficiaries	- Near real-time monitoring of AESIs - Regular observed-to-expected analyses conducted using comparison groups including historical rates - Other comparisons include concurrent unvaccinated and concurrent vaccinated intervals	- Access to all military beneficiaries' medical records - Medical records are robust - COVID-19 vaccination uptake data updated daily	- Electronic health record coding challenges - Incomplete data when care is not sought or received outside of the military health system
Newly established systems for COVID-19 vaccine safety monitoring and evaluation						
v-safe	CDC	Voluntary smartphone-based system that used text messaging to invite completion of web-based surveys	V-safe enrollees (or parents, guardians, caregivers) who completed surveys across the United States	- Active surveillance for reactogenicity and health impact events following COVID-19 vaccine - Identifies pregnancy status at and near the time of COVID-19 vaccination - Longitudinal data - Approximately 10 million people registered in v-safe	- Provides timely data on reactogenic events and what to expect following vaccination - Quickly collects national data, including information on those who experienced no symptoms or health impacts	- Design optimized for smartphone users - Lack of generalizability - Convenience sample of vaccinated persons - Incomplete reporting with decreasing survey participation over time
IHS NPTC	IHS	Email-based notification system surveying potential vaccine-related AEs at sentinel sites	AI/AN populations	- 61 sentinel sites representing 12 IHS areas	- Bi-weekly analyses of data - Serves as a method for early identification of AEs - Active outreach to IHS programs - Supports co-reporting to VAERS	- Fluctuating sentinel site participation in the biweekly survey
COVID-19 Vaccine Pregnancy Registry	CDC	- Participant-reported data captured through interviews - Medical record data on select participants	Pregnant persons consented and enrolled in the registry	- Pregnant persons identified through v-safe - Approximately 23,000 pregnant persons enrolled	- Early source of safety data on maternal and infant outcomes following COVID-19 primary series vaccines	- Data are collected from participant-reported interviews - Challenges adjudicating complex clinical outcomes - Variable availability and completeness of medical record data - Generalizability of findings limited
Myocarditis Outcomes After mRNA COVID-19 Vaccination Study	CDC	- Verified myocarditis cases reported to VAERS	Patients (and their healthcare providers)	- Approximately 800 case patients identified for surveillance; 62% participated in the study - Most patients were considered recovered at least 90 days after onset of myocarditis following mRNA COVID-19 vaccination	- Provided information on longer-term outcomes of myocarditis following mRNA COVID-19 vaccination - Included patient interview data, medical record data, and healthcare provider interview data	- Comparison of clinical courses among patients was challenging due to lack of available clinical practice guidelines - Selection bias as cases were identified through VAERS - Absence of a control group

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Table 2 (continued)

System	Managing federal agency	Data source	Population under surveillance	Characteristics	Strengths specific to COVID-19 vaccine safety monitoring	Limitations specific to COVID-19 vaccine safety monitoring
Clinical consultation						
CISA	CDC	Medical information from healthcare providers about their patients and from healthcare records included in reports to VAERS	N/A	- In-depth review of complex cases - Input and feedback from vaccine safety experts to provide guidance about vaccine decision-making to healthcare providers for their patients or technical expertise for vaccine safety issues and surveillance - CISA also conducts clinical research on COVID-19 vaccine safety	- U.S. healthcare providers with a complex vaccine safety question about a specific patient may contact CISA to request a consultation - Vaccine safety experts and medical specialists with expertise in areas of adult and pediatric medicine are available to support consultations and vaccine safety monitoring efforts	- Cases are selected for the purpose of providing technical assistance and clinical guidance to U.S. healthcare providers; CISA does not constitute a primary surveillance system

Abbreviations: AE, adverse event; AESI, adverse events of special interest; AI/AN, American Indian and Alaska Native; BEST, Biologics Effectiveness and Safety System; CDC, Centers for Disease Control and Prevention; CISA, Clinical Immunization Safety Assessment Project; CMS, Centers for Medicare and Medicaid Services; DHA-IHD, Defense Health Agency Immunization Healthcare Division; DoD, Department of Defense; DMSS, Defense Medical Surveillance System; EHR, electronic health record; FDA, U.S. Food and Drug Administration; IHS, Indian Health Service; IHS VAERS, Indian Health Service Vaccine Adverse Event Reporting System; IHS NPTC, Indian Health Service National Pharmacy and Therapeutics Committee; IIS, Immunization Information System; I-STAR, Indian Health Service Safety Tracking and Response system; VA, Department of Veterans Affairs; VA ADERS, Veterans Affairs Adverse Drug Event Reporting System; VAERS, Vaccine Adverse Event Reporting System; VSD, Vaccine Safety Datalink.

voluntary self-identification of reports originating from IHS federal, tribal, and Urban Indian Organization facilities. This functionality supported the analysis of COVID-19 vaccine safety data in tribal communities served by the IHS system of care. Through May 2023, over 1400 such reports have been identified. COVID-19 vaccine safety data observed among Native Americans was consistent with safety data from other populations within VAERS and other U.S. vaccine safety surveillance systems.

2.3. Indian Health Service Safety Tracking and Response System (I-STAR)

The IHS I-STAR system is a passive safety surveillance system deployed at all IHS federal healthcare facilities and a few participating tribal sites. I-STAR provided a platform for the early detection of healthcare worker and patient COVID-19 vaccine AEs as well as vaccine administration errors. Through June 1, 2023, I-STAR received 151 reports. I-STAR integrates logistics to encourage and support reporting to VAERS. I-STAR is subject to the limitations of passive surveillance. Because I-STAR is primarily deployed at federal sites, it had limited utility for the collection of worker-related vaccine AEs from tribal and Urban Indian Organization locations.

2.4. Department of Veterans Affairs Adverse Event Reporting System (VA ADERS)

The VA implemented a voluntary, internet-based AE reporting system called VA ADERS in 2007 [10]. This system collects reports of AEs potentially related to medications and vaccines into a centralized database. Healthcare providers can report AEs that occur within or outside the VA healthcare system to the VA ADERS database using a standardized template. Additional information may be included from the VA's EHR Vaccine AE reports submitted via the VA ADERS form.

COVID-19 AEs experienced by vaccinated VA employees and patients were reported using the VA ADERS system from December 2020 until June 2023. Weekly reports of COVID-19 vaccine AEs were summarized to include demographic information, serious and non-serious report status, and reporting rates of 11 pre-specified AESIs by vaccine type and dose number, and the top 10 reported symptoms. The most common AEs reported to VA ADERS for each vaccine were local

injection site and mild systemic reactions. A total of 10,846 reports concerning veterans were received through July 2023, after administration of 11.76 million vaccine doses. VA ADERS is subject to the limitations of passive surveillance.

2.5. DoD Vaccine Adverse Event Reporting System

U.S. military surveillance for AEs following COVID-19 vaccination is led by DHA IHD [27]. Clinicians in this organization have experience and expertise in evaluating AEs following vaccinations that are of special interest to the military, such as AEs following anthrax and smallpox vaccines [28]. DHA IHD clinicians receive weekly copies of VAERS reports from CDC which indicate military affiliation or association. Reports are linked to medical records and case follow-up is provided when clinically appropriate [29].

Prior to COVID-19, DHA IHD received an average of 1100 VAERS reports annually. In 2021, due to COVID-19 vaccine AE reporting requirements mandated by FDA, the military received and reviewed 7722 VAERS reports. These reports included an early safety signal for myocarditis following mRNA COVID-19 vaccination [30]. AESI cases were validated through medical record review. Reports also included a substantial number of vaccination errors not associated with adverse health events, consistent with EUA reporting requirements to VAERS [31]. By late 2022, the number of VAERS submissions received by DHA IHD had returned to pre-pandemic levels.

3. Active Surveillance

There are two basic types of active surveillance for vaccine safety. The first is near real-time sequential monitoring in EHR or administrative/claims-based systems using data generated for direct healthcare services. These data are not necessarily collected for public health surveillance or research purposes and often must be further formatted and structured for vaccine safety analyses. Although active surveillance in population-based data systems has advantages over passive surveillance, statistical associations often rely on automated analyses and may require complementary medical chart review of ICD-10 coded diagnoses for more definitive analyses. These statistical associations do not establish causality, and require further assessment, which may include formal epidemiologic studies. Another limitation of

surveillance using ICD-10 coded diagnoses is the variable nature of the codes in identifying true incident cases.

The second type of active surveillance involves active outreach and primary data collection, such as enrollment of vaccinated persons into a safety surveillance project and administering individual surveys to evaluate post-vaccination experiences. In this type of active surveillance, the patient or parent/caregiver of the patient are active participants in the safety monitoring process and are aware that they are participating in a public health surveillance activity.

3.1. Vaccine Safety Datalink (VSD)

The VSD, established in 1990, is a collaborative effort between CDC's Immunization Safety Office and thirteen integrated health-care systems [32]. The primary function of VSD is to conduct vaccine safety surveillance and research to help guide public health and regulatory action for U.S. vaccines and to inform policymakers and the public on vaccine safety [12]. VSD's estimated annual enrolled population is approximately 15.5 million persons, with an annual birth cohort of approximately 115,000 births. VSD uses EHR data which contains information on patient demographics, vaccinations, health outcomes, and other medical information from healthcare encounters [32]. In addition to EHR data, VSD also captures COVID-19 vaccinations from state immunization information systems (IIS) and through data-sharing agreements with private pharmacies. Since 2004, VSD has been conducting near real-time sequential safety monitoring, known as Rapid Cycle Analysis (RCA), which analyzes data that are updated weekly [12,33]. For COVID-19 vaccine safety RCA, VSD used data from 8 sites with a population of approximately 12.5 million persons to analyze up to 23 pre-specified ICD-10 coded outcomes (depending on age of the vaccinated person and the vaccine received). COVID-19 RCA monitoring assessed most outcomes using a novel vaccinated concurrent comparator method which compared concurrent (i.e., on the same day) outcome incidence in vaccinated individuals in a risk interval closer in time to vaccination (days 1–21) with an outcome incidence in vaccinated individuals in a comparison interval further in time from vaccination (days 22–42). Sequential statistical testing used a pre-specified threshold to detect safety signals. Vaccinated concurrent comparator analyses were adjusted for age, sex, race/ethnicity, VSD site and calendar date, which minimized potential confounding and biases that can impact alternative methods [34]. Medical record review was conducted to verify medical diagnoses and timing of symptom onset for acute disseminated encephalomyelitis, anaphylaxis, cerebral venous sinus thrombosis, GBS, ischemic stroke, myocarditis/pericarditis, thrombosis with thrombocytopenia syndrome, and transverse myelitis. VSD also separately conducted near real-time surveillance on the safety of COVID-19 vaccination in pregnant people which assessed acute maternal outcomes, pregnancy outcomes, and infant outcomes.

VSD COVID-19 vaccine safety RCA monitoring detected an increased risk for myocarditis and pericarditis in selected age strata following primary series and monovalent booster vaccination with both mRNA COVID-19 vaccines [34–36]. VSD RCA also identified a statistically higher rate of GBS following Ad26.CoV2.S vaccination when compared to the rate after mRNA COVID-19 vaccines; no evidence of an increased risk of GBS was identified following the mRNA COVID-19 vaccines [37]. RCA monitoring detected statistical signals for acute myocardial infarction (AMI) and venous thromboembolism (VTE) after different combinations of primary series mRNA COVID-19 vaccination [38]. Similar findings in FDA real-time surveillance for AMI and pulmonary embolism (PE), which is a complication of VTE, were further assessed by FDA using self-controlled methods (see section on Near Real-Time Surveillance in Centers for Medicare & Medicaid Services data). A statistical signal for ischemic stroke was detected following BNT162b2 bivalent COVID-19 vaccination among persons ages ≥ 65 years. Many of the stroke cases in this age group had also received simultaneous high-dose seasonal influenza vaccine in VSD RCA monitoring [39]. As the bivalent

COVID-19 vaccination program progressed and more data accumulated, the signal attenuated. Factors other than vaccination (e.g., unmeasured confounding) may have contributed to the initial statistical finding; CDC and FDA are conducting additional epidemiologic analyses. Finally, VSD found no safety concerns for either medically attended acute AEs or adverse pregnancy outcomes such as spontaneous abortion and stillbirth among pregnant people who received COVID-19 vaccination [40,41].

The VSD's COVID-19 RCA provided timely near real-time signal detection and signal assessment capability in a well-defined population. The limitations of VSD include its predominantly insured population in specific U.S. regions, which may limit generalizability of findings to the broader U.S. population. Additionally, VSD may be underpowered to detect very rare outcomes, despite its large population size.

3.2. Biologics Effectiveness and Safety System (BEST)

The FDA BEST Initiative was launched in 2017 as an active surveillance program for biologic products regulated by the Center for Biologics Evaluation and Research (CBER), Office of Biostatistics and Pharmacovigilance (OBPV) [13]. For COVID-19 vaccine safety monitoring, FDA has used BEST to monitor COVID-19 vaccine safety using administrative claims systems covering approximately 75 million health plan beneficiaries from CVS Health, Optum (pre-adjudicated claims), and Celeron. BEST's infrastructure includes analytic methods and tools development, including the ability to rapidly develop custom statistical analysis programs, and a network of partners who provide external subject matter experts to support CBER's vaccine safety surveillance efforts. Administrative claims data are linked to IIS data from approximately 30 states and jurisdictions allowing for comprehensive capture of COVID-19 vaccine administrations. The data are representative of privately insured individuals and cover persons from all parts of the United States and of all ages including the pediatric population and persons ages 18–64 years.

Following emergency use authorization of the first COVID-19 vaccines in the United States, FDA began conducting Near Real-Time Surveillance (NRS). Analyses of up to 19 AESIs using ICD-coded diagnoses, depending on age and vaccine, occurred on a bi-monthly or monthly schedule [42,43]. BEST employed a comparator based on background AESI rates for 2019, prior to the pandemic. The duration of risk periods evaluated post-vaccination periods, and risk thresholds for statistical signaling were pre-specified by outcome [43]. Among persons ages 12–64 years, anaphylaxis and myocarditis/pericarditis outcomes (in males ages 12–30 years) were detected as statistical signals and determined to be associated with vaccination [42]. The limitations of NRS are that it is a rapid screening method with limited control of confounding, given the comparator used [44]. Accordingly, statistical signal results do not establish causality and are further evaluated through a rigorous, more fully adjusted epidemiological study.

3.3. Near Real-time Surveillance in Centers for Medicare & Medicaid Services Data

Since 2005, FDA CBER has collaborated with the CMS to monitor and evaluate the safety and effectiveness of vaccines and other biologics in the 60 million individuals enrolled annually in Medicare fee-for-service programs, which includes many individuals aged ≥ 65 years in the United States. FDA and CMS conducted NRS for 16 ICD-10 coded AESIs on a weekly basis for all COVID-19 vaccines using approaches similar to BEST monitoring. Results for persons ages ≥ 65 years identified statistical signals following mRNA COVID-19 vaccination for anaphylaxis, PE, AMI, immune thrombocytopenia (ITP), and disseminated intravascular coagulation (DIC) [43]. FDA conducted rigorous self-controlled analyses and medical chart review of PE, AMI, ITP, and DIC cases from the study in the CMS database; the PE outcome was not consistently observed in follow-up studies and the other three AESIs signals were not confirmed to be associated with COVID-19 vaccination [44].

3.4. Department of Veterans Affairs Vaccine Rapid Cycle Analysis

VA's vaccine active surveillance program was initially developed in 2010 to monitor the safety of the 2009 H1N1 pandemic influenza vaccine and became fully operational in 2012. In December 2020, VA expanded its active surveillance program to monitor the safety of COVID-19 vaccines for approximately 9 million eligible veterans enrolled in the VA healthcare system. Prespecified ICD-10 coded AESIs were identified primarily from inpatient and emergency visits for most outcomes, with a small number from outpatient visits. Historical comparators were patients who received any vaccination (e.g., influenza, herpes zoster, hepatitis) during the 5 years preceding the COVID-19 vaccination period in the United States. The length of the risk window used depended on the AESI (i.e., primary risk window 1–21 days for all outcomes except for anaphylaxis [1 day]). A Poisson-based maximized sequential probability test (MaxSPRT) was used as the VA's primary RCA test statistic [11], adjusted for age and sex. Any safety signals identified through VA RCA or by other federal agencies (e.g., myocarditis/pericarditis) were evaluated through chart reviews conducted by the VA's pharmacovigilance personnel [15]. The only statistical signal identified by the VA RCA was anaphylaxis after mRNA COVID-19 vaccines; this finding was not unexpected based on results from other safety systems. The VA also conducted a comprehensive safety evaluation of mortality following the mRNA COVID-19 vaccines, in which no difference in mortality was identified when comparing unvaccinated and vaccinated patients during a short evaluation period; decreased mortality among the vaccinated population was identified during a longer evaluation (60 days). (Internal Communication, manuscript submitted).

The VA active surveillance system continues to be used for evaluation of the safety of updated COVID-19 vaccines and other routine vaccines. A limitation of including all veteran patients who received a COVID-19 vaccine is the possibility that certain outcomes were not captured for those veterans who use another system for their primary healthcare. Another limitation is the demographic characteristics of patients treated in VA (i.e., skewing older and male, and potentially less healthy than the general population), which may limit the ability to optimally evaluate vaccine safety in younger and female veteran subgroups.

3.5. Defense Medical Surveillance System (DMSS)

The U.S. military has a robust EHR system that is incorporated in the DMSS used to conduct disease surveillance and monitor vaccine and drug safety [16]. This system has been leveraged to ensure complete case-finding for vaccine safety when passive surveillance systems suggest a possible safety signal [45]. Past reviews of vaccine safety data have demonstrated strong concordance between DMSS data and VAERS reports [46].

During the COVID-19 vaccination program, DMSS conducted active surveillance of 26 AESIs among military beneficiaries. Daily, DMSS verified uptake of vaccinations among different demographic groups [47] and during different periods of vaccine availability [48]. Rates of ICD-10 coded diagnoses of AESIs after COVID-19 vaccination were compared to those of these diagnoses among unvaccinated service members, when applicable, and to pre-pandemic rates. Self-controlled methods were also used to monitor AESIs. Except for verifying the safety signal associated with cardiac AEs (i.e., myocarditis and pericarditis after mRNA COVID-19 vaccination), DMSS did not detect any other statistical associations. Limitations include incomplete data when care is not sought by the patient or when care is received outside of the military health system.

4. Case Consultations

4.1. Clinical Immunization Safety Assessment Project (CISA)

CDC's CISA Project is a collaboration of vaccine safety experts from the CDC's Immunization Safety Office, seven academic medical centers, subject matter experts, and other federal partners. CISA's primary goal is to serve as a vaccine safety resource for U.S. healthcare providers with complex vaccine safety questions about specific patients and to assist in immunization decision-making [17]. For the most complicated vaccine safety cases, particularly in situations where published guidance from CDC, the Advisory Committee on Immunization Practices (ACIP) and professional medical groups is limited, CISA uses a structured and confidential case review process.

As part of the CDC COVID-19 emergency response effort, CISA enhanced its clinical consultation service to accommodate an increased number of consultations and include subject matter experts in a wide range of adult and pediatric subspecialties. Specifically, the clinical service expanded to provide 24-hour/7-day coverage. Subsequently, CDC received and triaged hundreds of vaccine safety clinical inquiries, with a corresponding increase in the structured consultations to assess the most complicated clinical inquiries from healthcare providers, including one published case report [49]. CISA experts were consulted and provided individual input on several vaccine safety topics of interest, including anaphylaxis following mRNA COVID-19 vaccines, TTS, myocarditis and pericarditis, and GBS [50–52]. CISA also contributed to surveillance and conducted clinical research on COVID-19 vaccine safety to address evidence gaps.

5. New COVID-19 Vaccine Safety Monitoring Efforts

5.1. V-safe

V-safe is a new monitoring system that used smartphone technology and internet-based surveys to enhance VAERS reporting and to conduct rapid active surveillance for reactogenicity and health impact events after COVID-19 vaccination [19]. V-safe collected data about authorized and licensed COVID-19 vaccines during December 14, 2020, through June 30, 2023. Participation was voluntary and required self-registration. Once registered, v-safe participants received invitations via text message to complete brief surveys designed to solicit information about symptoms and health impacts experienced after vaccination. Reports of symptoms or health events associated with receipt of healthcare triggered a telephone call from a call center. If contacted, these participants were asked if they were interested in completing a VAERS report. More detailed descriptions of the implementation of v-safe, data collection and analysis, enhancements, and strengths and limitations are available [19].

More than 10.1 million people registered in v-safe through May 19, 2023, when enrollment closed. Of these, more than 9.5 million completed at least one health survey and were considered active v-safe participants; participants completed more than 151 million health surveys. CDC described and published data on symptoms and health events reported to v-safe and to VAERS after each ACIP recommendation for use of COVID-19 vaccines in specific populations [53–64]. The v-safe surveillance data indicated that local and systemic reactions reported after vaccination in large, diverse vaccinated U.S. populations were similar in frequency and severity to those noted in the pre-authorization clinical trials. These data provided clinicians with additional useful information for counseling their patients about common reactions expected after vaccination.

An important strength of v-safe is its ability to quickly collect safety data on a national scale, including reports from participants who experienced no symptoms or health events, a capability previously not available in U.S. vaccine safety monitoring. Limitations of v-safe for COVID-19 vaccination include a potential lack of generalizability due to

convenience sampling of vaccinated persons, incomplete reporting of events associated with decreasing survey participation with increasing time since vaccination, and the need for access to a smartphone to complete surveys [19].

5.2. CDC COVID-19 Vaccine Pregnancy Registry

The pre-authorization clinical trials for COVID-19 vaccines did not enroll pregnant persons. To monitor vaccine safety in this special population, CDC developed the CDC COVID-19 Vaccine Pregnancy Registry (registry) [21]. The registry capitalized on the availability of v-safe participants who indicated on the surveys that they were pregnant at the time of vaccination or conceived shortly after vaccination. The registry attempted to contact and enroll those participants who received a COVID-19 vaccine during pregnancy or up to 30 days prior to their last menstrual period. Depending on timing of enrollment relative to pregnancy end, enrolled participants were contacted up to 6 times (once during each trimester, approximately 1–2 months after the end of pregnancy, and for live births only, when the infant was at least three months old, and approximately 16–18 months after the end of pregnancy, whether a live birth or other outcome). Structured telephone interviews were conducted and data collected which included demographics, medical history, pregnancy complications and outcomes, and postpartum and infant health. Medical records of participants and infants were requested to assist with confirmation or elaboration of participant-reported information particularly for clinically complex outcomes such as birth defects, where standardized interview questions did not collect enough information to accurately classify the outcome.

Pregnant participants who reported vaccinations to v-safe from December 2020 through mid-June 2021, were potentially eligible for enrollment; over 23,000 participants were enrolled into the registry from January 2021 through August 2022 (data collection and analyses are ongoing). A detailed description of the registry methods, enrollment, strengths, and limitations are available elsewhere [19,65].

Limitations of the registry included participant-reported data from interviews, challenges adjudicating some of the more complex clinical outcomes, and variability in completeness of medical records (i.e., prenatal care, delivery hospital, and infant primary care). The generalizability of findings may be limited by the voluntary nature of both v-safe and the registry and the propensity for enrollment of most registry participants to be from the early period of the COVID-19 vaccination program. Questions of a more general nature were used to elicit information that would be capable of identifying potential safety issues with use of a novel vaccine administered on a population-wide basis. This approach resulted in some limitations in comprehensiveness of interview data, necessitating additional time and resources for acquisition and review of medical records to accurately classify some outcomes. In addition, the registry lacked an unvaccinated control group, thus outcome rates had to be compared to available background rates. Rates of some pregnancy and infant outcomes during the pandemic may be different when compared to pre-COVID-19 historical data because of direct effects of COVID-19 disease or changes in healthcare utilization during the pandemic.

Despite these limitations, the registry provided early data [66,67] to inform public health policy and clinical recommendations [68,69]. Preliminary analyses of data on spontaneous abortion (pregnancy loss < 20 weeks' gestation) [67] and analyses of other registry data [70,71] were considered during the development of recommendations for COVID-19 vaccination of pregnant people throughout pregnancy, as increased risks of SARS-CoV-2 infection during pregnancy were becoming more apparent [72–75]. Through July 2023, no safety signals have been detected in the registry data.

5.3. Indian Health Service National Pharmacy & Therapeutics Committee (NPTC)

The IHS NPTC [20] conducted sentinel site surveillance to assist early identification of potential COVID-19 vaccine safety concerns in the tribal communities served by this IHS system of care. Initially implemented to support agency allocation and distribution of remdesivir during the early months of the pandemic, in December 2020 COVID-19 vaccine safety questions were added to the biweekly NPTC Emerging Treatments Survey (NETS) conducted by the IHS NPTC Operations Team. Sixty-one initial sentinel sites, including ambulatory care clinics and hospitals, were identified to represent each of the 12 IHS Areas and including federal, tribal, and Urban Indian Organization (UIO) programs. By January 2023, participation was expanded to include over 280 IHS (federal), Tribal, and Urban Indian Organization program locations receiving COVID-19 medical countermeasures from the IHS National Supply Service Center.

Initially conducted by telephone, NPTC sentinel site surveillance evolved into a biweekly e-mail to sites requesting information related to vaccine AEs and to support co-reporting to VAERS. NETS surveillance for COVID-19 vaccine safety included similar information related to the utilization and safety of other COVID-19 medical countermeasures. Information was collected primarily by pharmacy personnel using a simple questionnaire. Following each analysis, biweekly reports were shared with key stakeholders, including the IHS COVID-19 Vaccine Task Force. NPTC sentinel site surveillance allows for active outreach to IHS programs, early identification of potential AEs, assistance for reporting to VAERS, and broad engagement with partners in the NPTC surveillance network, including participation from federal, tribal, and urban programs. One important limitation was variable response rates from sites to the biweekly surveys, which averaged between 45 and 60 %.

6. Case Example: Myocarditis and pericarditis following mRNA COVID-19 vaccination

Myocarditis and pericarditis were not identified as safety concerns in pre-authorization clinical trials of mRNA COVID-19 vaccines [76]. Initial findings of a potential safety concern for myocarditis and pericarditis associated with mRNA COVID-19 vaccination emerged in 2021 from reports from Israel and DHA IHD. Israel had quickly achieved high primary series BNT162b2 vaccination coverage among its population by spring of 2021 [77,78]. Post-implementation safety monitoring and evaluation in Israel detected an unexpectedly high number of myocarditis cases following COVID-19 vaccination, especially in adolescent and young adult males following dose 2 of the primary series [79,80]. In the United States, the first indication of a safety signal for myocarditis was detected in the U.S. military population also in the spring of 2021 [30]. Through DHA IHD monitoring of military VAERS reports, health officials noticed an unexpectedly high number of post-COVID-19 vaccination myocarditis cases in younger male servicemembers [30].

Israeli and DHA IHD health officials communicated their findings to CDC and briefed the ACIP Vaccine Safety Technical (VaST) work group [81], a work group established specifically to review and interpret COVID-19 vaccine safety data. As a result, CDC and FDA increased their vigilance in monitoring for myocarditis and pericarditis following mRNA COVID-19 vaccination, worked with specialists to develop standardized case definitions for these conditions [50], and engaged experts from CDC's CISA project to assist with evaluating case reports to VAERS and informing clinical guidance [52]. Data from available sources consistently demonstrated higher-than-expected reports of myocarditis following mRNA COVID-19 vaccination, most often in adolescent and young adult males following dose 2 of the primary vaccination series. Time to symptom onset in these individuals clustered within several days after vaccination. Reporting rates, especially after dose 2, were substantially higher than expected based on background rates of myocarditis in some age and sex strata [77–80,82].

During the spring of 2021 CDC and FDA were conducting active surveillance for cardiac outcomes through their respective systems, as previously described. Although VSD had not yet detected a statistical signal for myocarditis/pericarditis associated with either of the mRNA COVID-19 vaccines, a stratified analysis using the vaccinated concurrent comparator methodology detected an elevated rate ratio among people ages 12–39 years for myocarditis/pericarditis associated with mRNA COVID-19 vaccination. Again, cases occurred predominantly among males following dose 2 of the primary series, with symptom onset clustering within several days after vaccination [34]. Additional VSD analyses indicated excess risk for myocarditis/pericarditis following both mRNA vaccines among people ages 18–39 years; most cases occurred in males and moderately higher risk was found after mRNA-1273 compared to BNT162b2 vaccination [36]. Findings in FDA's BEST System also were consistent [83]. Therefore, the body of evidence from multiple U.S. and global monitoring systems was sufficient to establish a causal relationship between mRNA COVID-19 vaccination and myocarditis among adolescents and young adults, particularly among males and after dose 2.

7. Follow-up of myocarditis outcomes reported to VAERS after mRNA COVID-19 vaccination

Initial reviews of myocarditis cases after mRNA COVID-19 vaccination reported to passive surveillance systems suggested that most patients experienced symptom resolution by the time of hospital discharge and had generally favorable outcomes, particularly when compared with patients with viral myocarditis. However, data on medium-term and longer-term prognoses were limited [82,84]. In August 2021, CDC began a follow-up survey of verified myocarditis cases reported to VAERS to describe outcomes in patients in the age group 12–29 years, those with the highest risk of myocarditis after mRNA COVID-19 vaccination [22]. Follow-up on patients at least 90 days after symptom onset was attempted for cases of myocarditis reported to VAERS during the period January 12 through November 5, 2021. In total more than 800 individuals were identified for this follow-up surveillance project [22]. The survey included a component administered to patients or parents/guardians to ascertain quality of life, previous medical history, need for ongoing medication for myocarditis, and presence of clinical symptoms in the two weeks prior to the date of the survey. A second component asked treating healthcare providers about their patient's cardiac health and functional status, including objective measures such as findings from electrocardiogram, echocardiogram, cardiac magnetic resonance imaging (cMRI), troponin levels, exercise stress testing, and ambulatory rhythm monitoring.

The initial patient follow-up surveys showed that 81% of patients were considered by their cardiologist or other healthcare provider to have either fully or probably fully recovered. Self-reported quality-of-life measures were also similar to pre-COVID-19 pandemic norms among adolescents and young adults. To better understand the outcomes of individuals not considered by their healthcare provider to be fully or probably fully recovered at least 90 days after symptom onset, CDC conducted telephone surveys of both patients and cardiologists or other healthcare providers to describe outcomes at least 1 year after symptom onset (data collection and analysis are ongoing).

FDA supported and collaborated on the MACiV (Myocarditis After COVID Vaccination) Multicenter Study, a large retrospective observational study, enrolling patients ≤ 30 years of age with a clinical diagnosis of acute myocarditis following COVID-19 vaccination from 38 sites, based on clinical presentation, abnormal biomarkers and/or cardiovascular imaging findings per CDC criteria [85].

8. Communication of findings

Sharing COVID-19 vaccine safety information and surveillance findings in a timely and transparent manner was critical for the success

of the U.S. national COVID-19 vaccination program. CDC and FDA posted COVID-19 vaccine safety monitoring protocols online [13,25] and updated the VaST and the ACIP COVID-19 workgroup regularly [8,81]. An objective of the VaST was to provide interpretation of COVID-19 vaccine safety surveillance findings in a timely and transparent manner for the COVID-19 ACIP working group and the public. VaST reports were publicly posted summarizing discussions and data interpretation from the various data systems at ACIP meetings [86]. COVID-19 vaccine safety data were regularly presented to ACIP [87], FDA's Vaccine and Related Biological Products Advisory Committee (VRBPAC) [88], and WHO's Global Advisory Committee on Vaccine Safety (GACVS) [89]. Important safety analyses from the federal monitoring systems were published on preprint servers, in CDC's Morbidity and Mortality Weekly Report, and in peer-reviewed biomedical journals to ensure timely dissemination of information. Specifically, CDC published preliminary safety findings from VAERS and v-safe as soon as possible after each updated COVID-19 vaccine recommendation. CISA provided input to CDC's Interim Clinical Considerations for Use of COVID-19 Vaccines, particularly as it related to anaphylaxis, TTS, and multi-system inflammatory syndrome in children and adults following vaccination [90]. Plain language, lay-friendly communications were made available to the general public on CDC's, DOD's, FDA's, and VA's COVID-19 webpages, on digital and social media channels, and to public and private partners [88,91]. CDC maintained a web page on its COVID-19 site to communicate timely information on selected AEs reported after COVID-19 vaccination [92]. This web page was updated weekly to ensure the public and healthcare providers had the most accurate and up-to-date information about AEs following COVID-19 vaccination. Through these diverse venues, experts and the public alike were kept informed on the safety of COVID-19 vaccines.

9. Discussion

During the U.S. COVID-19 pandemic vaccination program, enhanced safety monitoring of vaccines and timely, accurate, and transparent communication of safety findings were essential components of the COVID-19 response. The regulatory and reporting requirements for federally purchased vaccines, expectations for timely and accurate data to inform policy decisions, and the coordinated actions of several U.S. government agencies and their surveillance partners culminated in the largest and most intense vaccine safety monitoring effort in U.S. history. Hundreds of millions of COVID-19 vaccinations were administered, new safety monitoring systems were introduced, and a new ACIP technical working group for COVID-19 vaccine safety was created. The planning phase for safety monitoring began in mid-2020 and involved substantial interagency consultation and collaboration on selection of surveillance outcomes, protocol and plan development, and development of new systems and processes. Continuous engagement between the federal agencies encouraged the exchange of ideas and information, including updates on findings from monitoring once vaccination began in December 2020. The U.S. vaccine safety monitoring infrastructure demonstrated its robustness and responsiveness; multiple systems managed by different federal agencies, using different data sources and methods, continuously monitored the safety of COVID-19 vaccines as new recommendations for age and risk groups were implemented. As described, this infrastructure included systems conducting passive surveillance, active surveillance, surveillance for pregnancy and infant outcomes, clinical consultations, and long-term follow-up on individual cases of myocarditis following vaccination. Publication of safety findings in scientific journals, presentations to federal advisory committees, postings on agency websites, and outreach through social media were venues to rapidly distribute the most up-to-date information to stakeholders and the public.

Some of the safety monitoring systems used during the COVID-19 vaccination program were well established (e.g., DHA IHD, VA RCA, VAERS, VSD, BEST [FDA Sentinel Initiative], and FDA's collaboration

with CMS for active surveillance). Existing systems were supplemented by enhanced follow-up and rapid clinical reviews to verify cases of AESIs and pre-specified surveillance outcomes. Accurate and timely data on COVID-19 vaccine doses administered resulting from jurisdictional reporting requirements, which were consolidated in CDC's COVID Data Tracker, greatly enhanced the value of passive surveillance beyond its typically numerator-only data [1]. A new program, v-safe, used innovative digital tools to add capabilities that did not exist prior to the pandemic response [19].

The coordinated efforts of the U.S. agencies involved in COVID-19 vaccine safety monitoring produced tangible results in the form of public health and regulatory actions that both informed the healthcare provider community and addressed public concerns. In Europe, COVID-19 vaccine safety data from country-specific reporting and through systems such as EudraVigilance were critical for informing the European Medicines Agency's Pharmacovigilance Risk Assessment Committee for safety assessments and risk management [93,94]. The Yellow Card System was used in the United Kingdom to inform regulatory and public health decisions [95].

The U.S. monitoring systems informed ACIP and VRBPAC benefit-risk analyses and supported regulatory decision-making for activities such as EUAs, licensures and approvals, safety related communications, and labeling changes [96]. Examples include the rapid detection of TTS following Ad26.CoV2.S vaccination and later the detection of an increased risk of GBS following Ad26.CoV2.S vaccination, which each led to updates to the Ad26.CoV2.S EUA documents and ultimately a preferential recommendation for mRNA COVID-19 vaccines over the Ad26.CoV2.S vaccine. Following the detection of the TTS signal, FDA and CDC recommended a pause in the use of Ad26.CoV2.S vaccine and CDC issued a CDC Health Alert Network (HAN) health alert to allow time for further evaluation and for notification of healthcare providers about the proper clinical evaluation and management of vaccine-associated TTS cases [51,97]. Following the detection of the myocarditis/pericarditis signal after mRNA vaccination, FDA updated the language in the relevant EUA documents and CDC updated its interim clinical considerations guidance for healthcare providers regarding the risks of myocarditis/pericarditis risk after mRNA COVID-19 vaccination [52,98].

In general, the findings from the comprehensive federal vaccine monitoring efforts have provided robust data supporting the safety of COVID-19 vaccines. After the administration and monitoring of more than 676 million doses in U.S. residents, serious AEs were rarely observed following vaccination. Timely and transparent communication of these findings were provided through regular updates to federal advisory committees, state health departments and other partners, and the American public [14,91]. The experience gained and lessons learned during the U.S. COVID-19 vaccine safety monitoring program have well positioned the United States and the federal agencies responsible for safety monitoring to respond to the next pandemic should rapid deployment of large volumes of vaccines to protect the public's health be necessary.

Disclaimer: The findings and conclusions of this report are those of the authors and do not necessarily represent the official positions of the Centers for Disease Control and Prevention, the U.S. Food and Drug Administration, the Department of Defense, the Department of Veterans Affairs, and the Indian Health Service.

Credit authorship contribution statement

Julianne Gee: Conceptualization, Visualization, Writing – original draft, Writing – review & editing. **Tom T. Shimabukuro:** Conceptualization, Writing – original draft, Writing – review & editing. **John R. Su:** Conceptualization, Writing – original draft, Writing – review & editing. **David Shay:** Writing – original draft, Writing – review & editing. **Margaret Ryan:** Writing – original draft, Writing – review & editing. **Sridhar V. Basavaraju:** Writing – review & editing. **Karen R. Broder:**

Writing – original draft, Writing – review & editing. **Matthew Clark:** Writing – review & editing. **C. Buddy Creech:** Writing – review & editing. **Francesca Cunningham:** Writing – original draft, Writing – review & editing. **Kristin Goddard:** Writing – review & editing. **Harrison Guy:** Writing – review & editing. **Kathryn M. Edwards:** **Richard Forshee:** Writing – original draft, Writing – review & editing. **Tanya Hamburger:** Writing – review & editing. **Anne M. Hause:** Writing – review & editing. **Nicola P. Klein:** Writing – review & editing. **Ian Kracalik:** Writing – original draft, Writing – review & editing. **Chris Lamer:** Writing – review & editing. **David A. Loran:** Writing – review & editing. **Michael M. McNeil:** Writing – review & editing. **Jay Montgomery:** Writing – review & editing. **Pedro Moro:** Writing – review & editing. **Tanya R. Myers:** Writing – review & editing. **Christine Olson:** Writing – original draft. **Matthew E. Oster:** Writing – review & editing. **Andrea J. Sharma:** Writing – original draft, Writing – review & editing. **Ryan Schupbach:** Writing – review & editing. **Eric Weintraub:** Writing – original draft, Writing – review & editing. **Brett Whitehead:** Writing – review & editing. **Steven Anderson:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Data availability

No data were used for the description of systems provided in the article.

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References

- [1] Centers for Disease Control and Prevention; Pages. Accessed at Centers for Disease Control and Prevention at https://covid.cdc.gov/covid-data-tracker/#vaccinations_vacc-people-booster-percent-total on June 1, 2023 2023.
- [2] Hlavka J, Rose, A. 2023; Pages. Accessed at Fortune Media IP Limited at <https://fortune.com/well/2023/05/16/how-much-did-covid-19-pandemic-coronavirus-cost-economy-14-trillion/> on June 1, 2023 2023.
- [3] Walmsley T, Rose A, John R, Wei D, Hlavka JP, Machado J, et al. Macroeconomic consequences of the COVID-19 pandemic. *Econ Model* 2023;120:106147.
- [4] Fitzpatrick M, Moghadas, SM, Pandey, A, Galvani, AP 2022; Pages. Accessed at The Commonwealth Fund at <https://www.commonwealthfund.org/blog/2022/two-years-covid-vaccines-prevented-millions-deaths-hospitalizations> on June 1, 2023 2023.
- [5] Dolgin E. The tangled history of mRNA vaccines. *Nature* 2021;597(7876):318–24.
- [6] Food and Drug Administration; Pages. Accessed at Food and Drug Administration at <https://www.fda.gov/media/151733/download> on June 3, 2023 2023.

- [7] Brighton Collaboration 2023;Pages. Accessed at Brighton Collaboration at <https://brightoncollaboration.us/category/pubs-tools/case-definitions/> on June 9, 2023 2023.
- [8] Council for International Organizations of Medical Sciences Workgroup VIII 2010; Pages. Accessed at Council for International Organizations of Medical Sciences at <https://cioms.ch/publications/product/practical-aspects-of-signal-detection-in-pharmacovigilance-report-of-cioms-working-group-viii/> on July 27, 2023 2023.
- [9] Shimabukuro TT, Nguyen M, Martin D, DeStefano F. Safety monitoring in the Vaccine Adverse Event Reporting System (VAERS). *Vaccine* 2015;33(36):4398–405.
- [10] Emmendorfer T, Glassman PA, Moore V, Leadholm TC, Good CB, Cunningham F. Monitoring adverse drug reactions across a nationwide health care system using information technology. *Am J Health Syst Pharm* 2012;69(4):321–8.
- [11] Kulldorff MDR, Kolczak M, Lewis E, Lieu T, Platt R A Maximized Sequential Probability Ratio Test for Drug and Vaccine Safety Surveillance. *Seq Anal* 2011;30(1):58–78.
- [12] Centers for Disease Control and Prevention 2022;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccinesafety/ensuringsafety/monitoring/vsd/index.html> on June 2, 2023 2023.
- [13] Food and Drug Administration 2023;Pages. Accessed at Food and Drug Administration at <https://bestinitiative.org/> on June 9, 2023 2023.
- [14] Food and Drug Administration 2021;Pages. Accessed at Food and Drug Administration at <https://www.fda.gov/vaccines-blood-biologics/safety-availability-biologics/covid-19-vaccine-safety-surveillance#CMS> on July 27, 2023 2023.
- [15] Lou J HK SC, Huang N, Pandey L, Thakkar B, Donahue M, Cunningham FE. Incidence rates and clinical characteristics of patients with confirmed myocarditis or pericarditis following COVID-19 mRNA vaccination: experience of the Veterans Health Administration through October 9, 2022. *Open Forum Infectious Diseases*. In Press.
- [16] DeFraites RF. The Armed Forces Health Surveillance Center: enhancing the Military Health System's public health capabilities. *BMC Public Health*. 2011;11 Suppl 2(Suppl 2):S1.
- [17] Centers for Disease Control and Prevention;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccinesafety/ensuringsafety/monitoring/cisa/index.html> on June 2, 2023 2023.
- [18] U.S. Government Accountability Office 2007;Pages. Accessed at U.S. Government Accountability Office at <https://www.gao.gov/products/gao-07-787r> on September 19, 2023 2023.
- [19] Myers TR, Marquez PL, Gee JM, Hause AM, Panagiotakopoulos L, Zhang B, et al. The v-safe after vaccination health checker: Active vaccine safety monitoring during CDC's COVID-19 pandemic response. *Vaccine* 2023;41(7):1310–8.
- [20] Indian Health Service 2023;Pages. Accessed at Indian Health Service at <https://www.ihs.gov/nptc/> on July 27, 2023 2023.
- [21] Centers for Disease Control and Prevention;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccinesafety/ensuringsafety/monitoring/v-safe/covid-preg-reg.html> on June 9, 2023 2023.
- [22] Kralcalk I, Oster ME, Broder KR, Cortese MM, Glover M, Shields K, et al. Outcomes at least 90 days since onset of myocarditis after mRNA COVID-19 vaccination in adolescents and young adults in the USA: a follow-up surveillance study. *Lancet Child Adolesc Health* 2022;6(11):788–98.
- [23] Food and Drug Administration 2023;Pages. Accessed at Food and Drug Administration at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrs.cfm?fr=312.32> on July 2, 2023 2023.
- [24] Centers for Disease Control and Prevention; Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccines/covid-19/vaccination-provider-support.html#provider-agreement> on May 31, 2023 2023.
- [25] Centers for Disease Control and Prevention;Pages. Accessed at Centers for Disease Control and Prevention at https://www.cdc.gov/vaccinesafety/ensuringsafety/monitoring/emergencypreparedness/index.html#anchor_1607961664745 on July 27, 2023 2023.
- [26] MacNeil JR, Su JR, Broder KR, Guh AY, Gargano JW, Wallace M, et al. Updated Recommendations from the Advisory Committee on Immunization Practices for Use of the Janssen (Johnson & Johnson) COVID-19 Vaccine After Reports of Thrombosis with Thrombocytopenia Syndrome Among Vaccine Recipients - United States, April 2021. *MMWR Morb Mortal Wkly Rep* 2021;70(17):651–6.
- [27] Martin BL, Nelson MR, Hershey JN, Engler RJ. Adverse reactions to vaccines. *Clin Rev Allergy Immunol* 2003;24(3):263–76.
- [28] McNeil MM, Cano M, E RM, Petersen BW, Engler RJ, Bryant-Genevieve MG. Ischemic cardiac events and other adverse events following ACAM2000(®) smallpox vaccine in the Vaccine Adverse Event Reporting System. *Vaccine*. 2014; 32(37):4758–65.
- [29] Bardenheier BH, Duderstadt SK, Engler RJ, McNeil MM. Adverse events following pandemic influenza A (H1N1) 2009 monovalent and seasonal influenza vaccinations during the 2009–2010 season in the active component U.S. military and civilians aged 17–44years reported to the Vaccine Adverse Event Reporting System. *Vaccine* 2016;34(37):4406–14.
- [30] Montgomery J, Ryan M, Engler R, Hoffman D, McClenathan B, Collins L, et al. Myocarditis following immunization with mRNA COVID-19 vaccines in members of the US Military. *JAMA Cardiol* 2021;6(10):1202–6.
- [31] Loran D, Reyes A, Angelo S, Carlungco D, Ryan M. COVID-19 Vaccination Errors: Challenges and Opportunities for Improving Support of Immunization Clinics. *Immunization Coalition Summit*. Riverside, CA; 2023.
- [32] McNeil MM, Gee J, Weintraub ES, Belongia EA, Lee GM, Glanz JM, et al. The Vaccine Safety Datalink: successes and challenges monitoring vaccine safety. *Vaccine* 2014;32(42):5390–8.
- [33] Lieu TA, Kulldorff M, Davis RL, Lewis EM, Weintraub E, Yih K, et al. Real-time vaccine safety surveillance for the early detection of adverse events. *Med Care*. 2007;45(10 Suppl 2):S89–95.
- [34] Klein NP, Lewis N, Goddard K, Fireman B, Zerbo O, Hanson KE, et al. Surveillance for adverse events after COVID-19 mRNA vaccination. *JAMA* 2021;326(14):1390–9.
- [35] Goddard K, Hanson KE, Lewis N, Weintraub E, Fireman B, Klein NP. Incidence of myocarditis/pericarditis following mRNA COVID-19 vaccination among children and younger adults in the United States. *Ann Intern Med* 2022;175(12):1169–771.
- [36] Goddard K, Lewis N, Fireman B, Weintraub E, Shimabukuro T, Zerbo O, et al. Risk of myocarditis and pericarditis following BNT162b2 and mRNA-1273 COVID-19 vaccination. *Vaccine* 2022;40(35):5153–9.
- [37] Hanson KE, Goddard K, Lewis N, Fireman B, Myers TR, Bakshi N, et al. Incidence of Guillain-Barré syndrome after COVID-19 vaccination in the vaccine safety datalink. *JAMA Netw Open* 2022;5(4):e228879.
- [38] Klein NP 2023;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccines/acip/meetings/downloads/slides-2023-09-12/07-COVID-Klein-508.pdf> on September 19, 2023 2023.
- [39] Klein NP, Shimabukuro, T 2023;Pages. Accessed at Food and Drug Administration at <https://www.fda.gov/media/164811/download> on July 27, 2023 2023.
- [40] DeSilva M, Haapala J, Vazquez-Benitez G, Vesco KK, Daley MF, Getahun D, et al. Evaluation of acute adverse events after Covid-19 vaccination during pregnancy. *N Engl J Med* 2022;387(2):187–9.
- [41] DeSilva MB, Haapala J, Vazquez-Benitez G, Boyce TG, Fuller CC, Daley MF, et al. Medically attended acute adverse events in pregnant people after coronavirus disease 2019 (COVID-19) booster vaccination. *Obstet Gynecol* 2023;142(1):125–9.
- [42] Lloyd PC, Hu M, Wong HL, Shoaibi A, Ke Zhou C, Lo AC, et al. Near real-time surveillance of safety outcomes in US COVID-19 vaccine recipients aged 12 to 64 years. *Vaccine* 2022;40(45):6481–8.
- [43] Wong HL, Tworokoski E, Ke Zhou C, Hu M, Thompson D, Lufkin B, et al. Surveillance of COVID-19 vaccine safety among elderly persons aged 65 years and older. *Vaccine* 2023;41(2):532–9.
- [44] Azadeh SLP, Wong HL, Clarke TC, Chillarige Y, Do R, Hu M, et al.; 2023. Accessed at medRxiv at <https://www.medrxiv.org/content/10.1101/2023.01.19.23284803v1> on June 9, 2023 2023.
- [45] Mandra AM, Superior MJ, Guagliardo SAJ, Hesse E, Pacha LA, Stidham RA, et al. Myopericarditis associated with smallpox vaccination among US Army personnel - fort hood, Texas, 2018. *Disaster Med Public Health Prep* 2022;16(3):1022–8.
- [46] McNeil MM, Ma G, Aranas A, Payne DC, Rose Jr CE. A comparative assessment of immunization records in the defense medical surveillance system and the vaccine adverse event reporting system. *Vaccine* 2007;25(17):3428–36.
- [47] Lang MA, Stahlman S, Wells NY, Fedgo AA, Patel DM, Chauhan A, et al. Disparities in COVID-19 vaccine initiation and completion among active component service members and health care personnel, 11 December 2020–12 March 2021. *Msmr*. 2021;28(4):2–9.
- [48] Winkler EL, Stahlman SL, Wells NY, Chauhan AV, Hibban KM, Costello AA, et al. COVID-19 Booster Vaccination in the U.S. Military, August 2021–January 2022. *Am J Prev Med*. 2023;64(2):270–4.
- [49] Hag K, Anyalechi EG, Schlaudecker EP, McKay R, Kamidani S, Manos CK, et al. Multiple MIS-C readmissions and giant coronary aneurysm after COVID-19 illness and vaccination: A case report. *Pediatr Infect Dis J* 2023;42(3):e64–9.
- [50] Gargano JW, Wallace M, Hadler SC, Langley G, Su JR, Oster ME, et al. Use of mRNA COVID-19 vaccine after reports of myocarditis among vaccine recipients: update from the advisory committee on immunization practices - United States, June 2021. *MMWR Morb Mortal Wkly Rep* 2021;70(27):977–82.
- [51] Centers for Disease Control and Prevention 2021;Pages. Accessed at CDC Health Alert Network at <https://emergency.cdc.gov/han/2021/han00442.asp> on June 2, 2023 2023.
- [52] Centers for Disease Control and Prevention; Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccines/covid-19/clinical-considerations/myocarditis.html> on May 30, 2023 2023.
- [53] Chapin-Bardales J, Gee J, Myers T. Reactogenicity Following Receipt of mRNA-Based COVID-19 Vaccines. *JAMA* 2021;325(21):2201–2.
- [54] Gee J, Marquez P, Su J, Calvert GM, Liu R, Myers T, et al. First Month of COVID-19 vaccine safety monitoring - United States, December 14, 2020–January 13, 2021. *MMWR Morb Mortal Wkly Rep* 2021;70(8):283–8.
- [55] Hause AM, Baggs J, Gee J, Marquez P, Myers TR, Shimabukuro TT, et al. Safety monitoring of an additional dose of COVID-19 vaccine - United States, August 12–September 19, 2021. *MMWR Morb Mortal Wkly Rep* 2021;70(39):1379–84.
- [56] Hause AM, Baggs J, Marquez P, Abara WE, Baumballt J, Blanc PG, et al. Safety monitoring of COVID-19 mRNA vaccine second booster doses among adults aged ≥50 years - United States, March 29, 2022–July 10, 2022. *MMWR Morb Mortal Wkly Rep* 2022;71(30):971–6.
- [57] Hause AM, Baggs J, Marquez P, Abara WE, Baumballt JG, Thompson D, et al. Safety monitoring of COVID-19 mRNA vaccine first booster doses among persons aged ≥12 years with presumed immunocompromise status - United States, January 12, 2022–March 28, 2022. *MMWR Morb Mortal Wkly Rep* 2022;71(28):899–903.
- [58] Hause AM, Baggs J, Marquez P, Abara WE, Olubajo B, Myers TR, et al. Safety monitoring of COVID-19 vaccine booster doses among persons aged 12–17 Years - United States, December 9, 2021–February 20, 2022. *MMWR Morb Mortal Wkly Rep* 2022;71(9):347–51.
- [59] Hause AM, Baggs J, Marquez P, Myers TR, Gee J, Su JR, et al. COVID-19 vaccine safety in children aged 5–11 years - United States, November 3–December 19, 2021. *MMWR Morb Mortal Wkly Rep* 2021;70(5152):1755–60.
- [60] Hause AM, Baggs J, Marquez P, Myers TR, Su JR, Huguely B, et al. Safety Monitoring of Pfizer-BioNTech COVID-19 Vaccine Booster Doses Among Children

- Aged 5–11 Years - United States, May 17–July 31, 2022. *MMWR Morb Mortal Wkly Rep* 2022;71(33):1047–51.
- [61] Hause AM, Gee J, Baggs J, Abara WE, Marquez P, Thompson D, et al. COVID-19 vaccine safety in adolescents aged 12–17 years - United States, December 14, 2020–July 16, 2021. *MMWR Morb Mortal Wkly Rep* 2021;70(31):1053–8.
 - [62] Hause AM, Marquez P, Zhang B, Myers TR, Gee J, Su JR, et al. Safety monitoring of bivalent COVID-19 mRNA vaccine booster doses among persons aged ≥12 years - United States, August 31–October 23, 2022. *MMWR Morb Mortal Wkly Rep* 2022; 71(44):1401–6.
 - [63] Hause AM, Marquez P, Zhang B, Myers TR, Gee J, Su JR, et al. COVID-19 mRNA vaccine safety among children aged 6 Months–5 Years - United States, June 18, 2022–August 21, 2022. *MMWR Morb Mortal Wkly Rep* 2022;71(35):1115–20.
 - [64] Hause AM, Marquez P, Zhang B, Su JR, Myers TR, Gee J, et al. Safety monitoring of Bivalent COVID-19 mRNA vaccine booster doses among children aged 5–11 years - United States, October 12–January 1, 2023. *MMWR Morb Mortal Wkly Rep* 2023; 72(2):39–43.
 - [65] Madni SA, Sharma AJ, Zauche LH, Waters AV, Nahabedian 3rd JF, Johnson T, Olson CK, CDC COVID-19 Vaccine Pregnancy Registry Work Group. CDC COVID-19 Vaccine Pregnancy Registry: Design, data collection, response rates, and cohort description. *Vaccine* 2024 Mar 7;42(7):1469–77. <https://doi.org/10.1016/j.vaccine.2023.11.061>. Epub 2023 Dec 6. PMID: 38057207.
 - [66] Shimabukuro TT, Kim SY, Myers TR, Moro PL, Oduyebo T, Panagiotakopoulos L, et al. Preliminary findings of mRNA Covid-19 vaccine safety in pregnant persons. *N Engl J Med* 2021;384(24):2273–82.
 - [67] Zauche LH, Wallace B, Smoots AN, Olson CK, Oduyebo T, Kim SY, et al. Receipt of mRNA Covid-19 vaccines and risk of spontaneous abortion. *N Engl J Med* 2021;385 (16):1533–5.
 - [68] American College of Obstetrics and Gynecology; Society for Maternal Fetal Medicine 2021;Pages. Accessed at American College of Obstetrics and Gynecology at <https://www.acog.org/news/news-releases/2021/07/acog-smfm-recommend-covid-19-vaccination-for-pregnant-individuals> on May 25, 2023 2023.
 - [69] Centers for Disease Control and Prevention 2021;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/media/releases/2021/s0811-vaccine-safe-pregnant.html> on May 25, 2023 2023.
 - [70] Ellington SR, Kharbanda, E.O., Olson, C.K., Fleming-Dutra, K.E. COVID-19 in pregnant people and infants ages 0-5 months; COVID-19 vaccine safety in pregnancy; Effectiveness of maternal COVID-19 vaccination in pregnant and infants. Advisory Committee on Immunization Practices. Atlanta, GA: Centers for Disease Control and Prevention; 2022.
 - [71] Olson CK. Updates from the v-safe COVID-19 vaccine pregnancy registry. Advisory Committee on Immunization Practices Atlanta, GA: Centers for Disease Control and Prevention; 2021.
 - [72] Jering KS, Claggett BL, Cunningham JW, Rosenthal N, Vardeny O, Greene MF, et al. Clinical characteristics and outcomes of hospitalized women giving birth with and without COVID-19. *JAMA Intern Med* 2021;181(5):714–7.
 - [73] Ko JY, DeSisto CL, Simeone RM, Ellington S, Galang RR, Oduyebo T, et al. Adverse pregnancy outcomes, maternal complications, and severe illness among US delivery hospitalizations with and without a coronavirus disease 2019 (COVID-19) diagnosis. *Clin Infect Dis* 2021;73(Suppl 1):S24–31.
 - [74] Prabhu M, Cagino K, Matthews KC, Friedlander RL, Glynn SM, Kubiak JM, et al. Pregnancy and postpartum outcomes in a universally tested population for SARS-CoV-2 in New York City: a prospective cohort study. *BJOG* 2020;127(12):1548–56.
 - [75] Zambrano LD, Ellington S, Strid P, Galang RR, Oduyebo T, Tong VT, et al. Update: characteristics of symptomatic women of reproductive age with laboratory-confirmed SARS-CoV-2 infection by pregnancy status - United States, January 22–October 3, 2020. *MMWR Morb Mortal Wkly Rep* 2020;69(44):1641–7.
 - [76] Food and Drug Administration 2020;Pages. Accessed at Food and Drug Administration at <https://web.archive.org/web/20201231184246/https://www.fda.gov/emergency-preparedness-and-response/coronavirus-disease-2019-covid-19/covid-19-vaccines> on May 23, 2023 2023.
 - [77] Leshem E, Wilder-Smith A. COVID-19 vaccine impact in Israel and a way out of the pandemic. *Lancet* 2021;397(10287):1783–5.
 - [78] Rosen B, Waitzberg R, Israeli A. Israel's rapid rollout of vaccinations for COVID-19. *Isr J Health Policy Res* 2021;10(1):6.
 - [79] Mevorach D, Anis E, Cedar N, Bromberg M, Haas EJ, Nadir E, et al. Myocarditis after BNT162b2 mRNA vaccine against Covid-19 in Israel. *N Engl J Med* 2021;385 (23):2140–9.
 - [80] Witberg G, Barda N, Hoss S, Richter I, Wiessman M, Aviv Y, et al. Myocarditis after Covid-19 vaccination in a large health care organization. *N Engl J Med* 2021;385 (23):2132–9.
 - [81] Markowitz LE, Hopkins Jr RH, Broder KR, Lee GM, Edwards KM, Daley MF, et al. COVID-19 Vaccine Safety Technical (VaST) Work Group: Enhancing vaccine safety monitoring during the pandemic. *Vaccine* 2024;42(S3):125549. <https://doi.org/10.1016/j.vaccine.2023.12.059>. S0264-410X(23)01505-0. Epub ahead of print. PMID: 38341293.
 - [82] Oster ME, Shay DK, Su JR, Gee J, Creech CB, Broder KR, et al. Myocarditis cases reported after mRNA-Based COVID-19 vaccination in the US From December 2020 to August 2021. *JAMA* 2022;327(4):331–40.
 - [83] Wong HL, Hu M, Zhou CK, Lloyd PC, Amend KL, Beachler DC, et al. Risk of myocarditis and pericarditis after the COVID-19 mRNA vaccination in the USA: a cohort study in claims databases. *Lancet* 2022;399(10342):2191–9.
 - [84] Bozkurt B, Kamat I, Hotez PJ. Myocarditis With COVID-19 mRNA Vaccines. *Circulation* 2021;144(6):471–84.
 - [85] Jain SS, Anderson SA, Steele J, et al. Outcomes in COVID-19 mRNA Vaccine-Associated Myocarditis in the Young in the United States: Results from The Myocarditis After COVID-19 Vaccination (MACiV) Multicenter Longitudinal Study. Available at SSRN: <https://ssrn.com/abstract=4681858> or <https://dx.doi.org/10.2139/ssrn.4681858>.
 - [86] Centers for Disease Control and Prevention 2022;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccines/acip/work-groups-vast/index.html> on July 2, 2023 2023.
 - [87] Centers for Disease Control and Prevention;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccines/acip/index.html> on June 1, 2023 2023.
 - [88] Food and Drug Administration 2023;Pages. Accessed at Food and Drug Administration at <https://www.fda.gov/advisory-committees/blood-vaccines-and-other-biologics/vaccines-and-related-biological-products-advisory-committee> on June 1, 2023 2023.
 - [89] World Health Organization 2023;Pages. Accessed at World Health Organization at <https://www.who.int/groups/global-advisory-committee-on-vaccine-safety> on June 4, 2023 2023.
 - [90] Centers for Disease Control and Prevention;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html#contraindications> on September 15, 2023 2023.
 - [91] Centers for Disease Control and Prevention;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/coronavirus/2019-ncov/index.html> on July 2, 2023 2023.
 - [92] Centers for Disease Control and Prevention;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/coronavirus/2019-ncov/vaccines/safety/safety-of-vaccines.html> on June 1, 2023 2023.
 - [93] European Medicines Agency. Accessed at European Medicines Agency at <https://www.ema.europa.eu/en/human-regulatory-overview/research-and-development/pharmacovigilance-research-and-development/eudravigilance> on February 11, 2024.
 - [94] European Medicines Agency. Accessed at European Medicines Agency at <https://www.ema.europa.eu/en/committees/pharmacovigilance-risk-assessment-committee-prac> on February 11, 2024.
 - [95] Medicines & Healthcare products Regulatory Agency. Accessed at Medicines & Healthcare products Regulatory Agency at <https://yellowcard.mhra.gov.uk/> on February 11, 2024.
 - [96] Funk PR, Yorgurtcu ON, Forshee RA, Anderson SA, Marks PW, Yang H. Benefit-risk assessment of COVID-19 vaccine, mRNA (Comirnaty) for age 16–29 years. *Vaccine* 2022;40(19):2781–9.
 - [97] Centers for Disease Control and Prevention 2021;Pages. Accessed at Centers for Disease Control and Prevention at <https://www.cdc.gov/media/releases/2021/s0413-JJ-vaccine.html> on September 29, 2023 2023.
 - [98] Food and Drug Administration 2021;Pages. Accessed at Food and Drug Administration at <https://www.fda.gov/news-events/press-announcements/coronavirus-covid-19-update-june-25-2021> on July 2, 2023 2023.